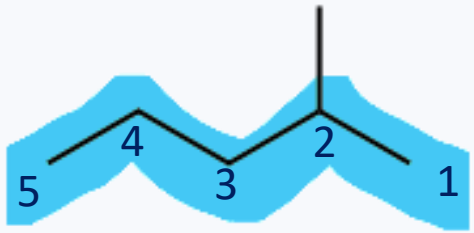
CARD A01 — STRUCTURE → NAME

Level: FOUNDATION Molecular formula (counted): C_6H_{14}

1. NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the end nearer the methyl branch.

2. NAMING TRAIL

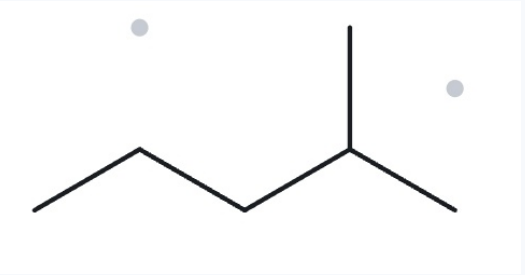
***Parent**: longest chain = 5 C → pentane.*
***Numbering**: left → right gives 4; right → left gives 2. The lower locant is 2, so number from the right.*
***Substituents**: methyl on C2 → 2-methyl.*

3. MY NAME (WRITTEN FIRST)

2-methylpentane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



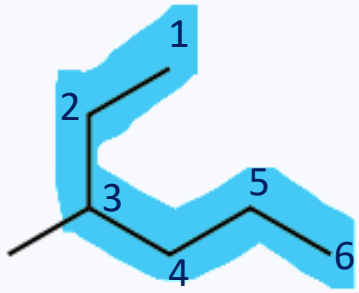
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A02 — STRUCTURE → NAME

Level: FOUNDATION Molecular formula (counted): C_7H_{16}

1. NUMBERED PARENT



Parent (6 C) highlighted; the parent chain follows the upper branch through the central junction and toward the right-hand end.

2. NAMING TRAIL

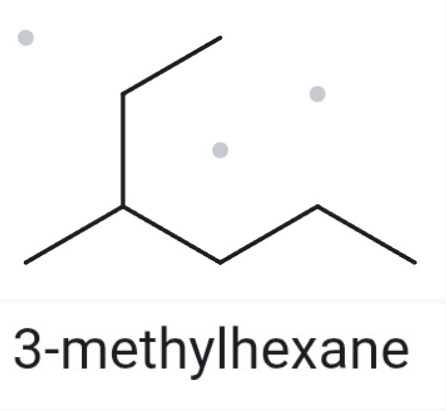
***Parent**: longest chain = 6 C → hexane..*
***Numbering**: one direction gives methyl on C4; the opposite direction gives methyl on C3. The lower locant is 3.*
***Substituents**: methyl on C3 → 3-methyl.*

3. MY NAME (WRITTEN FIRST)

3-methylhexane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



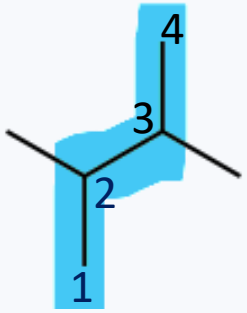
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A03 — STRUCTURE → NAME

Level: FOUNDATION Molecular formula (counted): C_6H_{14}

1. NUMBERED PARENT



Parent (4 C) highlighted; carbons numbered from either end because the structure is symmetrical.

2. NAMING TRAIL

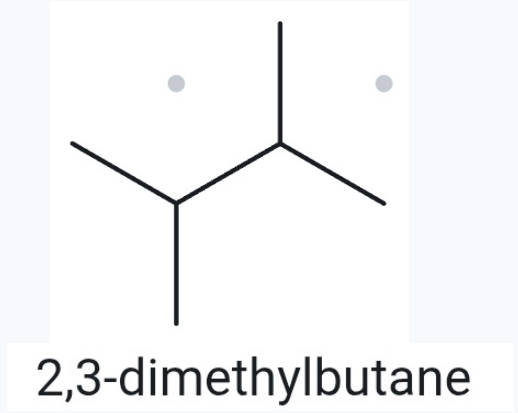
***Parent**: longest chain = 4 C → butane.*
***Numbering**: left → right gives 2,3; right → left also gives 2,3. Both directions produce the same locants.*
***Substituents**: methyl on C2 and C3 → 2,3-dimethyl.*

3. MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



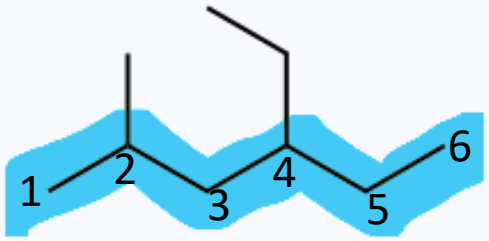
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A04 — STRUCTURE → NAME

Level: DEVELOPING Molecular formula (counted): C_9H_{20}

1. NUMBERED PARENT



Parent (6 C) highlighted; carbons numbered from the left-hand end to obtain the lower set of locants.

2. NAMING TRAIL

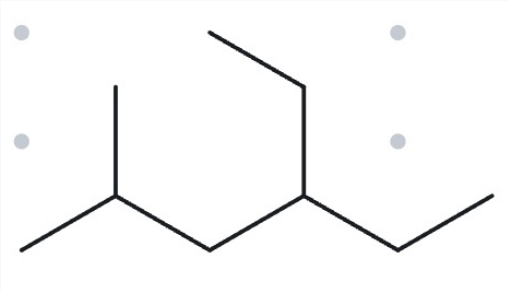
***Parent**: longest chain = 6 C → hexane.*
***Numbering**: left → right gives 2,4; right → left gives 3,5. The first point of difference is 2 versus 3, so number from the left.*
***Substituents**: ethyl on C4 and methyl on C2. Alphabetically, ethyl comes before methyl → 4-ethyl-2-methyl.*

3. MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



4-ethyl-2-methylhexane

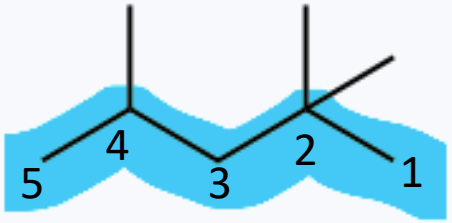
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A05 — STRUCTURE → NAME

Level: DEVELOPING Molecular formula (counted): C_8H_{18}

1. NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the right-hand end to give the lowest locants.

2. NAMING TRAIL

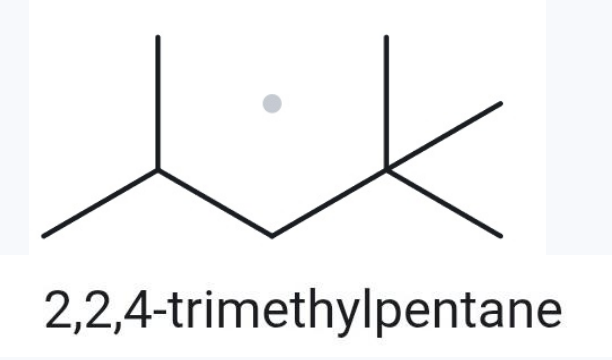
***Parent**: longest chain = 5 C → pentane.*
***Numbering**: left → right gives 2,4,4; right → left gives 2,2,4. The first point of difference is 2 versus 4, so number from the right.*
***Substituents**: methyl on C2, C2 and C4 → 2,2,4-trimethyl.*

3. MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



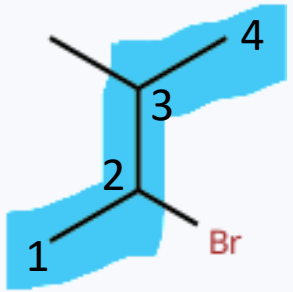
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A06 — STRUCTURE → NAME

Level: DEVELOPING Molecular formula (counted): $C_5H_{11}Br$

1. NUMBERED PARENT



Parent (4 C) highlighted; carbons numbered from the end that gives bromo the lower locant in the tie.

2. NAMING TRAIL

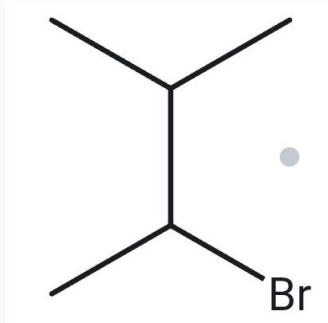
***Parent**: longest chain = 4 C → butane.*
***Numbering**: both directions give the locant set 2,3. Since the sets tie, the substituent cited first alphabetically receives the lower number.*
***Substituents**: bromo on C2 and methyl on C3 → 2-bromo-3-methyl.*

3. MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



2-bromo-3-methylbutane

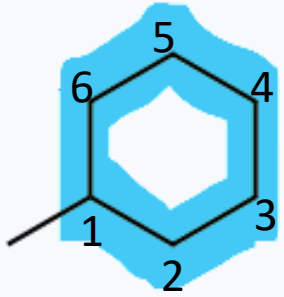
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A07 — STRUCTURE → NAME

Level: DEVELOPING Molecular formula (counted): C_7H_{14}

1. NUMBERED PARENT



Parent (6 C) highlighted as a cyclohexane ring; the attached methyl group is on C1. Numbering is unnecessary for a ring with only one substituent.

2. NAMING TRAIL

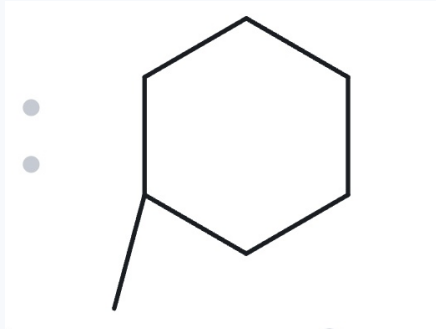
***Parent**: ring has 6 C → cyclohexane.*
***Numbering**: only one substituent is present, so no locant is required.*
***Substituents**: methyl attached to the ring → methylcyclohexane.*

3. MY NAME (WRITTEN FIRST)

methylcyclohexane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



methylcyclohexane

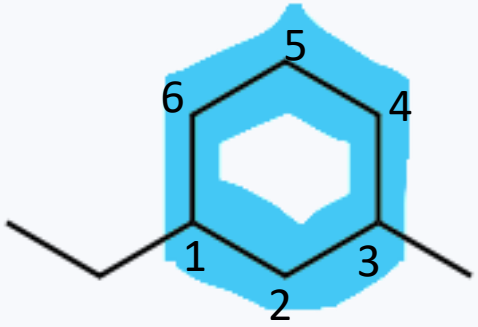
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A08 — STRUCTURE → NAME

Level: CHALLENGE Molecular formula (counted): C_9H_{18}

1. NUMBERED PARENT



Parent (6 C) highlighted as a cyclohexane ring; numbering starts at the carbon attached to the ethyl group.

2. NAMING TRAIL

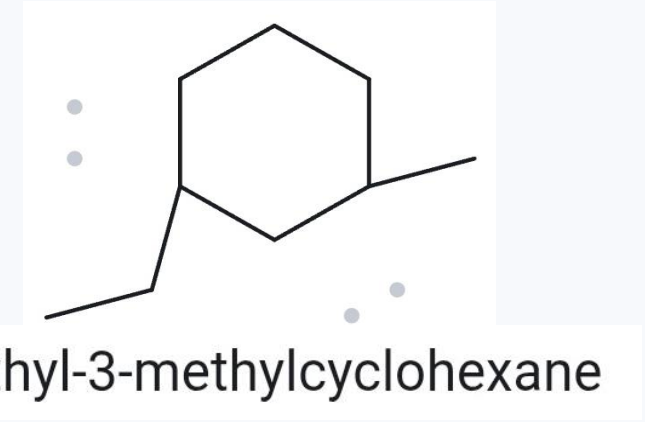
***Parent**: ring has 6 C → cyclohexane.*
***Numbering**: both directions give locants 1,3. Since the locant sets tie, ethyl receives C1 because it comes before methyl alphabetically.*
***Substituents**: ethyl on C1 and methyl on C3 → 1-ethyl-3-methyl.*

3. MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK

✓matched -The names I typed agree with the names returned by The Alkane, confirming that the structures and IUPAC naming rules were applied consistently.



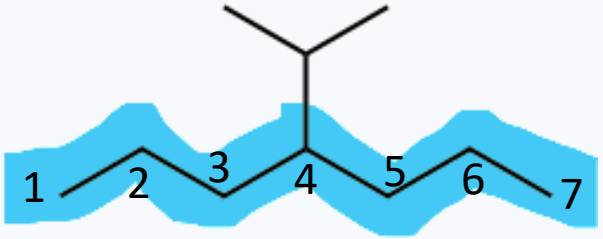
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A09 — STRUCTURE → NAME

Level: CHALLENGE Molecular formula (counted): $C_{10}H_{22}$

1. NUMBERED PARENT



Parent (7 C) highlighted; heptane chain selected as the longest continuous chain.

2. NAMING TRAIL

***Parent**: longest chain = 7 C → heptane. The upper branch is not included entirely in the parent chain because the longest continuous chain contains 7 carbons.*

***Numbering**: left → right gives 4; right → left also gives 4. Both directions produce the same locant.*

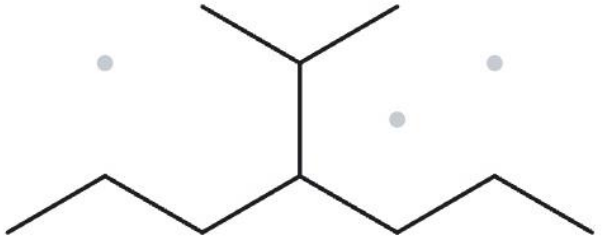
***Substituents**: propan-2-yl (isopropyl) on C4.*

3. MY NAME (WRITTEN FIRST)

4-isopropylheptane

4. THE ALKANE CHECK

✗ mismatched - The Alkane selected a six-carbon parent chain (hexane), but I selected the longest continuous seven-carbon chain (heptane) as the parent, so the resulting IUPAC names are different.



2-methyl-3-propylhexane

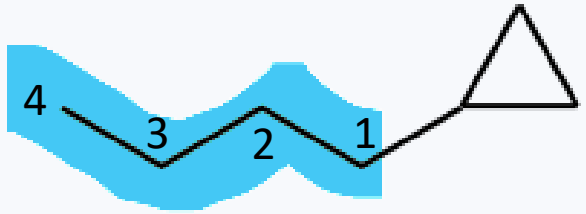
VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A10 — STRUCTURE → NAME

Level: _____ Molecular formula (counted): _____

1. NUMBERED PARENT



Parent (4 C) highlighted; butane chain selected because it has more carbons than the cyclopropane ring (3 C).

2. NAMING TRAIL

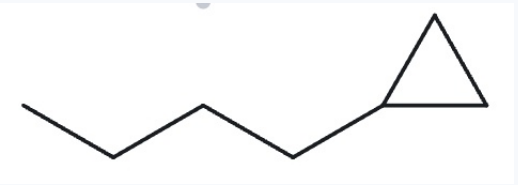
***Parent**: longest chain = 4 C → butane. The open chain has more carbons than the cyclopropane ring.*
***Numbering**: number from the end nearest the cyclopropyl substituent → substituent on C1.*
***Substituents**: cyclopropyl on C1.*

3. MY NAME (WRITTEN FIRST)

1-cyclopropylbutane

4. THE ALKANE CHECK

✗ mismatched - The Alkane selected the ring as the parent and named the open chain as a butyl substituent, while I selected the longer four-carbon open chain as the parent and named the ring as a cyclopropyl substituent.



butylcyclopropane

VERIFICATION STRIP

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

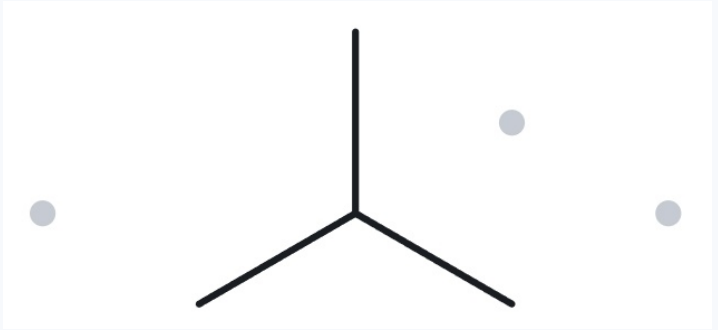
CARD B01 — NAME → STRUCTURE

Level: FOUNDATION Given name: 2-methylpropane

1. NAME DECODED

***Parent**: propane → 3 carbons*
***Substituents**: methyl at C-2*
***Expected Formula**: 3 + 1 = 4 C; open chain → C_4H_{10}.*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_4H_{10}.*
***Open chain**: C_nH_{2n+2} = C_4H_{10} ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

2-methylpropane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

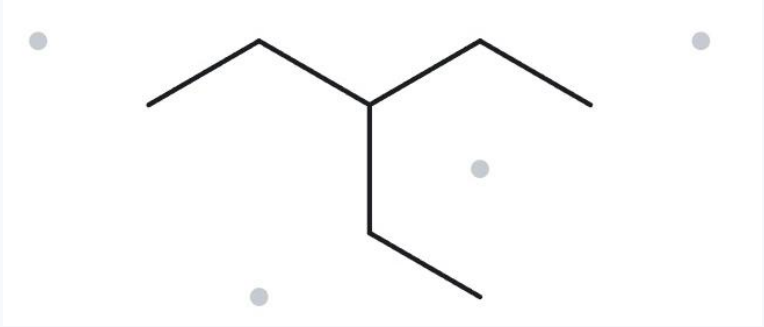
CARD B02 — NAME → STRUCTURE

Level: FOUNDATION Given name: 3-ethylpentane

1. NAME DECODED

***Parent**: pentane → 5-carbon chain*
***Substituents**: ethyl on C3*
***Expected Formula**: 5 + 2 = 7 C; open chain → C_7H_{16}*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_7H_{16}*
***Open chain**: $C_nH_{2n+2} = C_7H_{16}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

3-ethylpentane

✓Same as the given name

VERIFICATION STRIP

☐✓ parent length ☐✓ substituent positions ☐✓ formula agrees ☐✓ back-check recorded

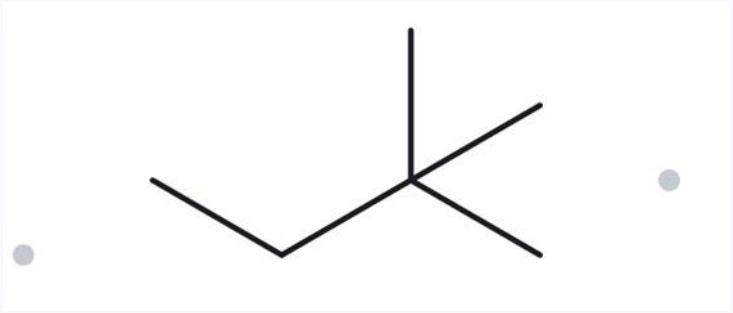
CARD B03 — NAME → STRUCTURE

Level: FOUNDATION Given name: 2,2-dimethylbutane

1. NAME DECODED

***Parent**: butane → 4-carbon chain.*
***Substituents**: two methyl groups on C2*
***Expected Formula**: 4 + 2 = 6 C; open chain → C_6H_{14}*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_6H_{14}*
***Open chain**: $C_nH_{2n+2} = C_6H_{14}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

2,2-dimethylbutane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

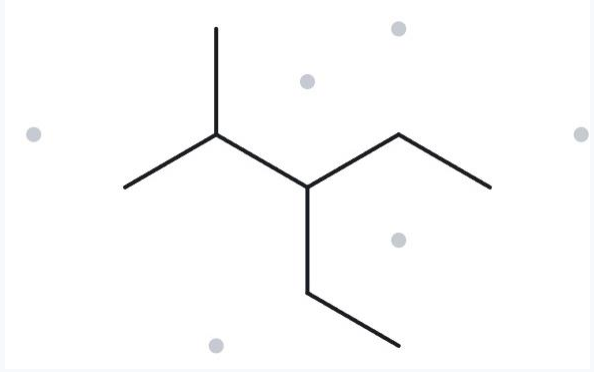
CARD B04 — NAME → STRUCTURE

Level: DEVELOPING Given name: 3-ethyl-2-methylpentane

1. NAME DECODED

***Parent**: pentane → 5-carbon chain*
***Substituents**: ethyl on C3; methyl on C2*
***Expected Formula**: 5 + 2 + 1 = 8 C; open chain → C_8H_{18}*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_8H_{18}*
***Open chain**: $C_nH_{2n+2} = C_8H_{18}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

3-ethyl-2-methylpentane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

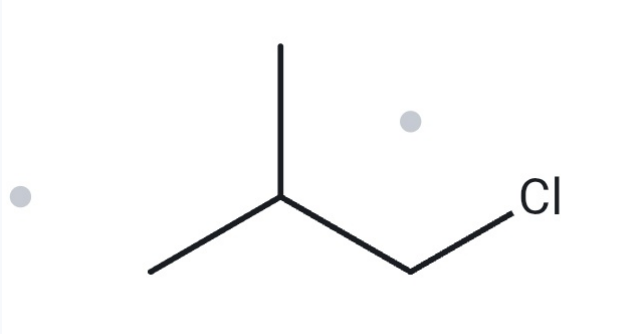
CARD B05 — NAME → STRUCTURE

Level: DEVELOPING Given name: 1-chloro-2-methylpropane

1. NAME DECODED

***Parent**: propane → 3 carbons*
***Substituents**: chloro on C1; methyl on C2*
***Expected Formula**: 3 + 1 = 4 C; open chain → C_4H_{10};*
Cl replaces one H → C_4H_9Cl

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_4H_9Cl*
***Open chain**: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

1-chloro-2-methylpropane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

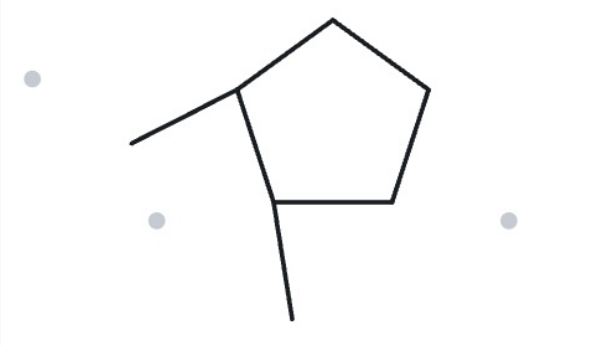
CARD B06 — NAME → STRUCTURE

Level: DEVELOPING Given name: 1,2-dimethylcyclopentane

1. NAME DECODED

***Parent**: cyclopentane → a 5-carbon ring*
***Substituents**: methyl on C1 and C2*
***Expected Formula**: 5 + 1 + 1 = 7 C; one ring → C_7H_{14}*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_7H_{14}*
***Open chain**: C_nH_{2n} = C_7H_{14} ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

1,2-dimethylcyclopentane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

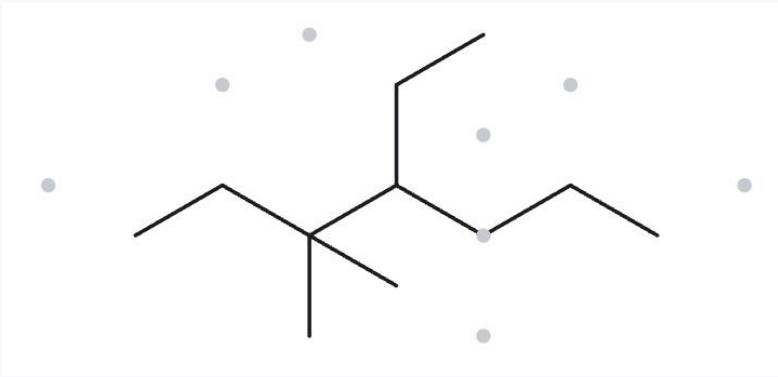
CARD B07 — NAME → STRUCTURE

Level: DEVELOPING Given name: 4-ethyl-3,3-dimethylheptane

1. NAME DECODED

Parent: *heptane → 7-carbon chain*
Substituents*: ethyl on C4; two methyl groups on C3*
Expected Formula*: 7 + 2 + 1 + 1 = 11 C; open chain →* $C_{11}H_{24}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing*:* $C_{11}H_{24}$
Open chain*:* $C_nH_{2n+2} = C_{11}H_{24}$ ✓

4. BACK-CHECK

Returned by The Alkane*:*

4-ethyl-3,3-dimethylheptane

✓***Same as the given name***

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

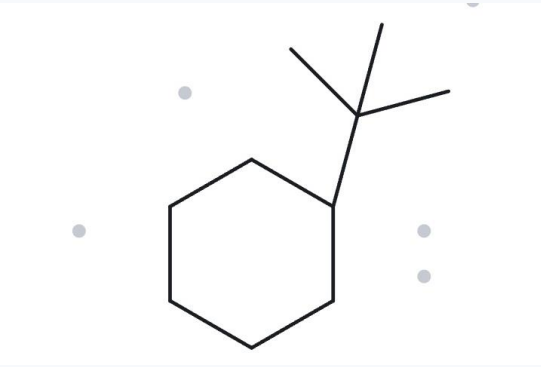
CARD B08 — NAME → STRUCTURE

Level: CHALLENGE Given name: tert-butylcyclohexane

1. NAME DECODED

***Parent**: cyclohexane → 6-carbon ring*
***Substituents**: tert-butyl on C1; the substituent contains 4 carbons*
***Expected Formula**: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: $C_{10}H_{20}$*
***Open chain**: C_nH_{2n} = $C_{10}H_{20}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

(1,1-dimethylethyl)cyclohexane

✓Same as the given name
(tert-butyl and 1,1-dimethylethyl are equivalent names for the same substituent.)

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

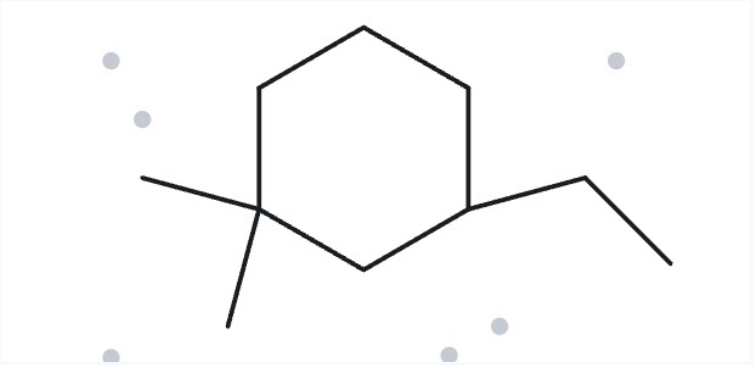
CARD B09 — NAME → STRUCTURE

Level: CHALLENGE Given name: 3-ethyl-1,1-dimethylcyclohexane

1. NAME DECODED

***Parent**: cyclohexane → 6-carbon ring*
***Substituents**: two methyl groups on C1; ethyl on C3*
***Expected Formula**: 6 + 1 + 1 + 2 = 10 C; one ring → $C_{10}H_{20}$*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: $C_{10}H_{20}$*
***Open chain**: C_nH_{2n} = $C_{10}H_{20}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

3-ethyl-1,1-dimethylcyclohexane

✓Same as the given name

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded

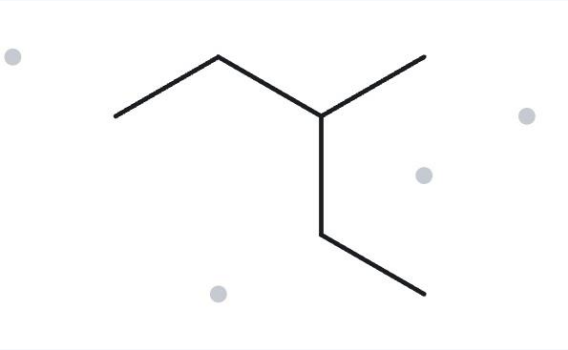
CARD B10 — NAME → STRUCTURE

Level: CHALLENGE Given name: 2-ethylbutane (deliberately incorrect name)

1. NAME DECODED

***Parent**: butane → a 4-carbon chain*
***Substituents**: ethyl on C2*
***Expected Formula**: 4 + 2 = 6 C; open chain → C_6H_{14}*

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

***Counted from the drawing**: C_6H_{14}*
***Open chain**: $C_nH_{2n+2} = C_6H_{14}$ ✓*

4. BACK-CHECK

***Returned by The Alkane**:*

3-methylpentane

***Correct name:** 3-methylpentane*
***Reason:** The longest continuous chain contains 5 carbon atoms, so pentane must be selected as the parent instead of butane*

VERIFICATION STRIP

☐ parent length ☐ substituent positions ☐ formula agrees ☐ back-check recorded